

Development Exams Exercise

Part I – Theory

(Total: 36 points) – Maximum Time: 30 minutes (+10)

Complete the answers to the 7 questions on this Google Form, using your ACT Google account.

GOOGLE FORMS URL: TBD (To be described)

As soon as you finish answering them you may proceed to the second part of the exams.

Part II – Implementation

(Total: 64 points) - Minimum Time: 90 minutes (+20)

Checkout your personalized copy of the **GIS** application from the SVN repository using your credentials:

SVN URL: TBD

You will find a main form for this application already designed, along with some incomplete class declarations. There are 3 data forms that can be opened from the main form. Each form handles a different kind of data inside the data store. Your personalized variation of the following general requirements and more details can be found under **requirements.pdf** file that lies in the root of your folder.

File: Requirements.pdf

MODULE 1 - CFormCountries

(24 points)

- 1.1 Create/use a class that implements the proper data structure that can keep **key-value pairs** for the capital cities of the world.
- 1.2 Instantiate an object for the data structure of 1.1, inside the constructor of the form, keeping the reference in a field. Load the data structure from the **countries.json** file, that is supplied inside the project folder. (*Adapt the CJSONFile class to load the file*).
- 1.3 Display the **list of keys sorted** in the text area of the form. Prefer the most efficient sorting algorithm from those that you have learned.

MODULE 2 - CFormContinents

(24 points)

2.1 Create a class that implements the proper data structure that can store the **parent-child relationships** that exist between continent and countries that belong to them.

2.2 Instantiate an object for the data structure of 2.1, inside the constructor of the form, keeping the reference in a field. Load the data structure from the lines of **continents-sample.txt**, that is supplied inside the project folder. *(Use the CTextFile class to load the file)*

- *(You need to understand how a parent-child relationship is encoded in the file)*
- *(You need to think how to properly create the data from a sequence of lines)*
- *(Consider using sorting to write a more easy loading sequence)*

2.3 Display the data structure in the text area of the form.

MODULE 3 - CFormHighways

(16 points)

3.1 Create a class that implements the proper data structure that can store cities that are **connected to each other** with highways.

3.2 Instantiate an object for the data structure of 3.1, inside the constructor of the form, keeping the reference in a field. Load the data structure from the **highways.txt**, that is supplied inside the project folder.

3.3 For each city **count the number of connections** to other cities. Create a secondary data structure to hold the city name and the number of connections.

3.4 Sort the cities by the count of connections to other cities and display their names in the text area of the form.